Nebius NVIDIA Hackathon Challenge

Britannica dataset: Historical Encyclopaedia Britannica scans.

Wikipedia dataset: Wikimedia structured contents.

Wikipedia access: Use the shared, pre-indexed API to search for candidate articles and retrieve their text. Follow the public Wikipedia API guide; you do not need to build this index.

Overview

Build a pipeline that matches historical Encyclopaedia Britannica entries to Wikipedia articles and compares how the two sources describe the same subjects.
Your use of all four Nemotron models listed below will be evaluated. The starter supplies setup code and model clients; your team builds the extraction workflow, matching, comparisons, agent tools, and website.

- Nemotron 3.5 Super VL
- Nemotron Parse 3.5
- Nemotron Embed VL
- Nemotron Rerank VL

Key question

Which Wikipedia articles best match the subjects in historical Britannica entries?

Getting Started

Github repository:
https://github.com/october-18-hackathon/nemotron-agentic-ai-challenge

Reminder: Nebius AI Cloud and Nebius Token Factory require separate sign-ups. You will need two accounts.

Use the participant setup guide to get started.

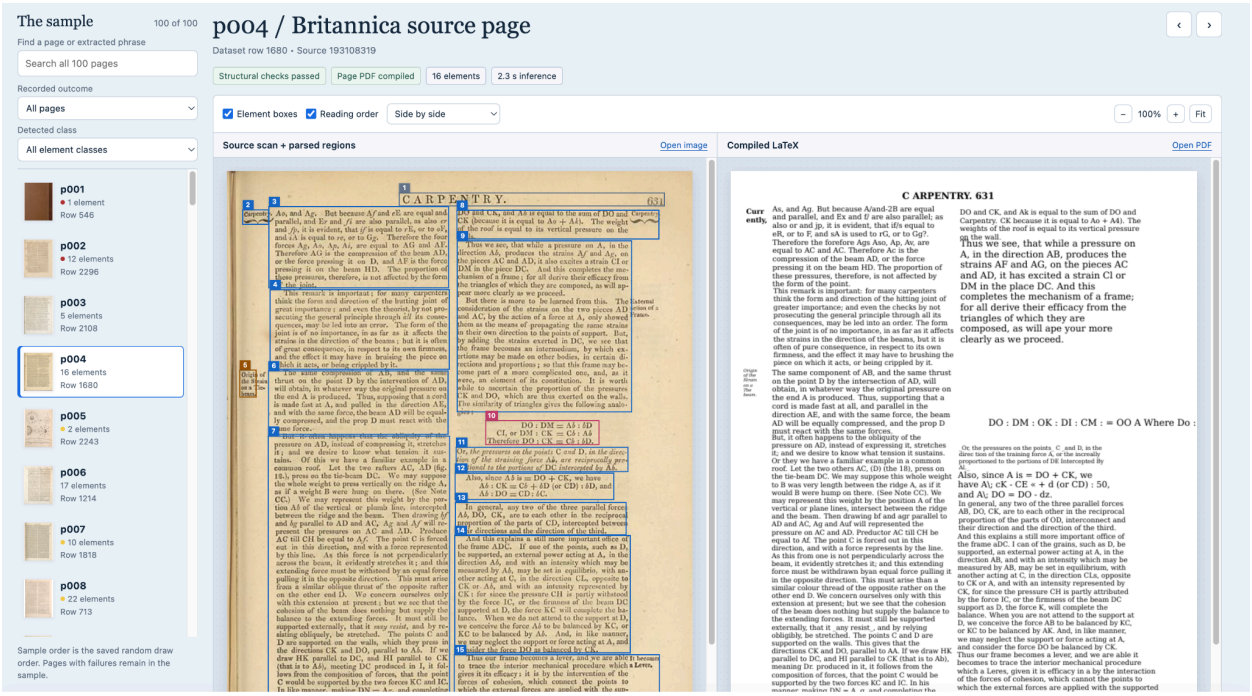
- **Nemotron Parse 3.5 (early access model) is pre-deployed and available through this endpoint: https://port8000-kh2kceraevwe0k1.tunnel.applications.uk-south2.nebius.cloud/docs**
- **Wikipedia is pre-indexed and available here** https://port8000-rks0xavparpzq59.tunnel.applications.uk-south2.nebius.cloud

Tasks

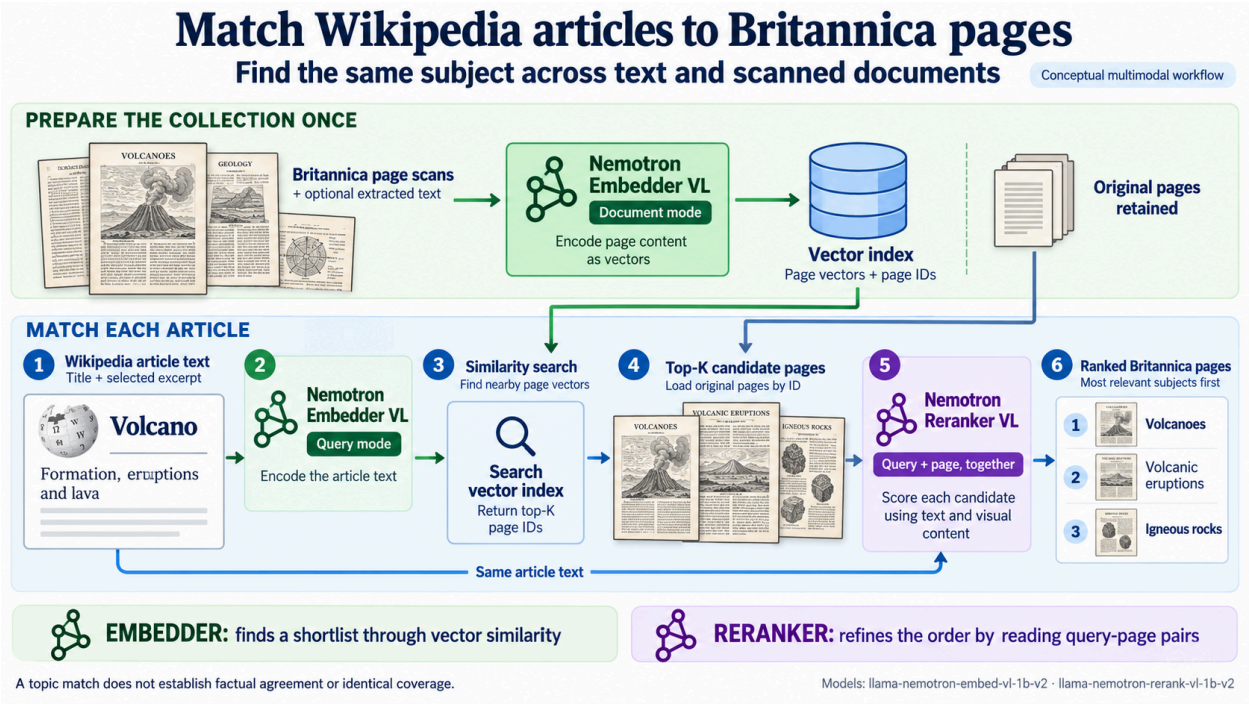
1. Extract the Britannica pages

    - Use Nemotron Parse 3.5 to extract content from your selected scans. Create a static website that presents the extracted Britannica content as compiled LaTeX documents.

Example archive view: a source scan with parsed regions alongside a compiled LaTeX document.



2. Match Britannica entries to Wikipedia articles



Select a sample of 100 historical Britannica pages, either at random or by hand. Find the Wikipedia articles that best match the entries in those pages. The starter provides a fixed sample with source checksums that you can use as a starting point.

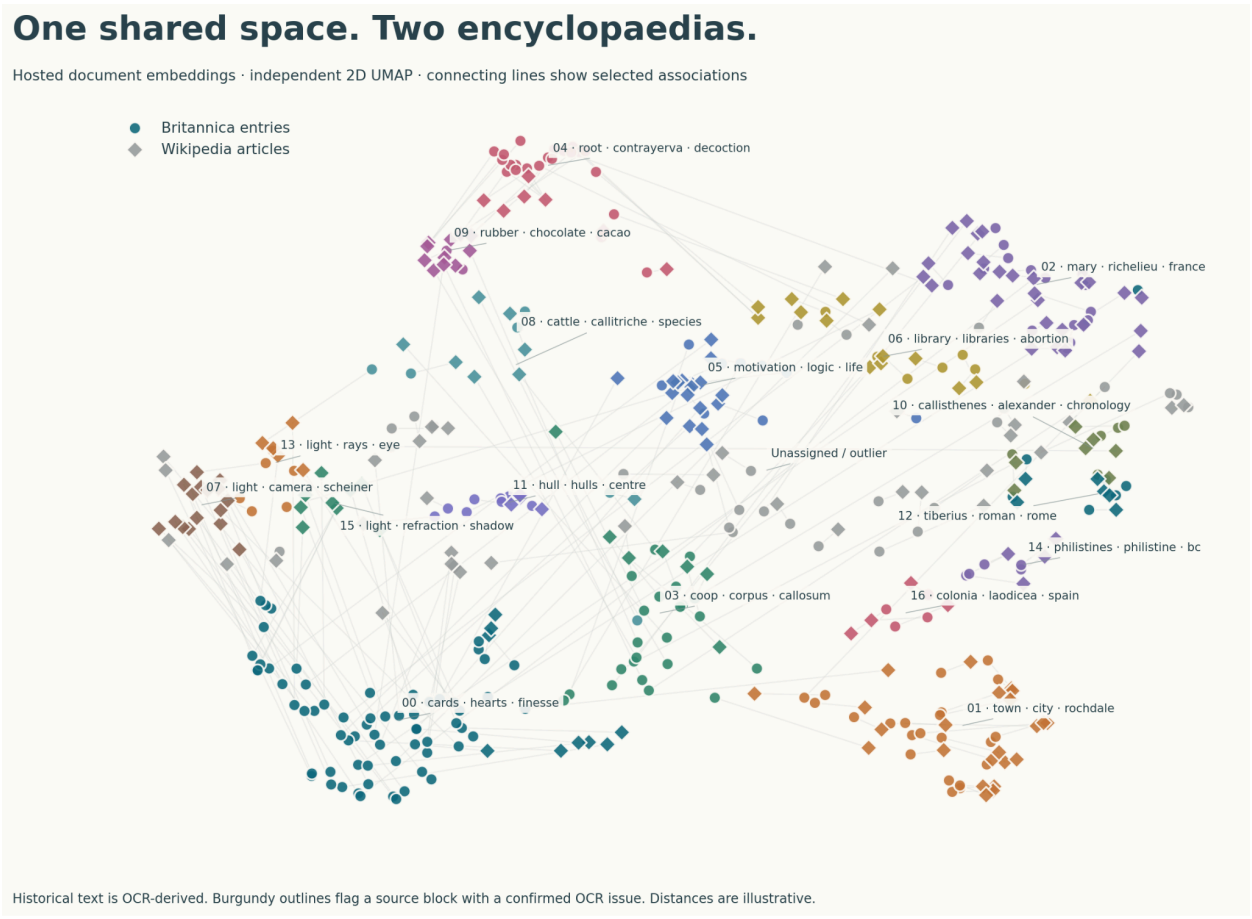
The diagram above illustrates the reverse direction, from Wikipedia text to Britannica scans. Your pipeline must start from Britannica entries and find Wikipedia articles. A relevance score does not establish factual agreement.

Public matching guidance: Follow the starter milestones and implement your own matching method.

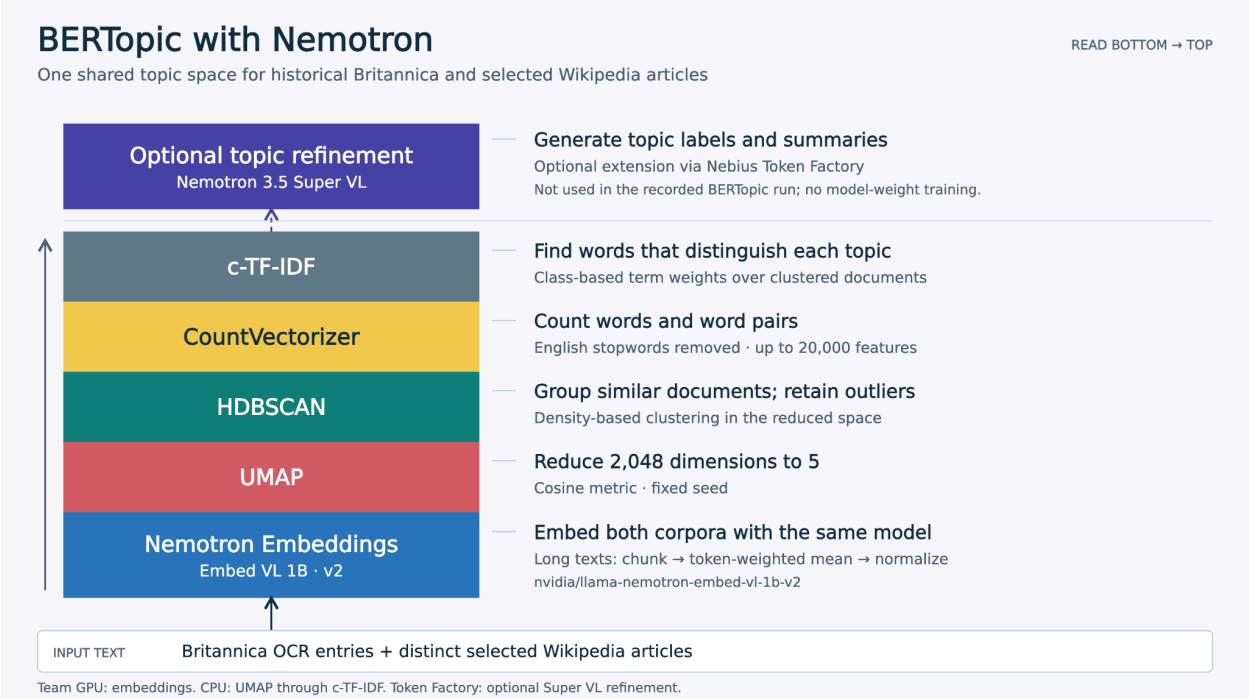
3. Compare matched articles

Document similarities and differences between matched entries and articles, such as vocabulary, word frequencies, or topics. Explain your method and cite evidence from both sources.

Example comparison output. Produce your own analysis from your selected sources.

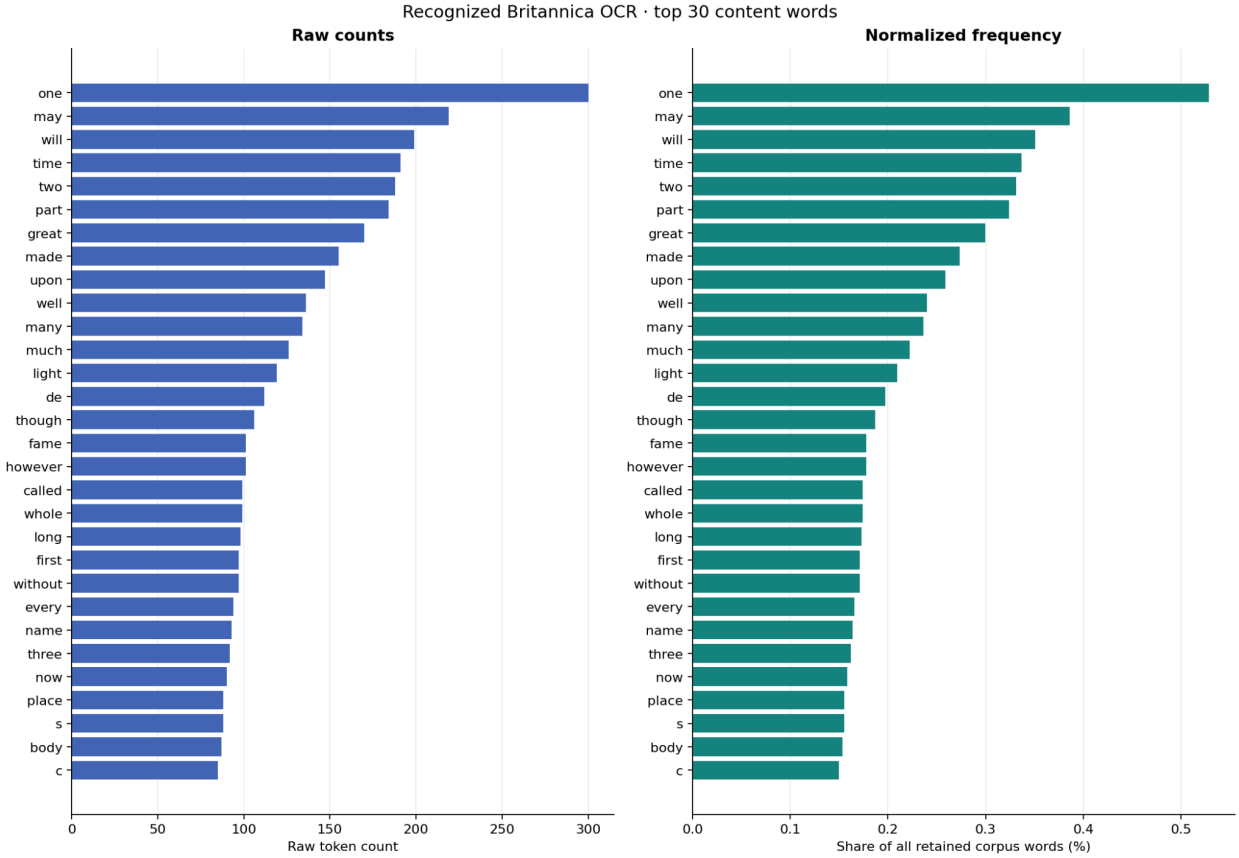


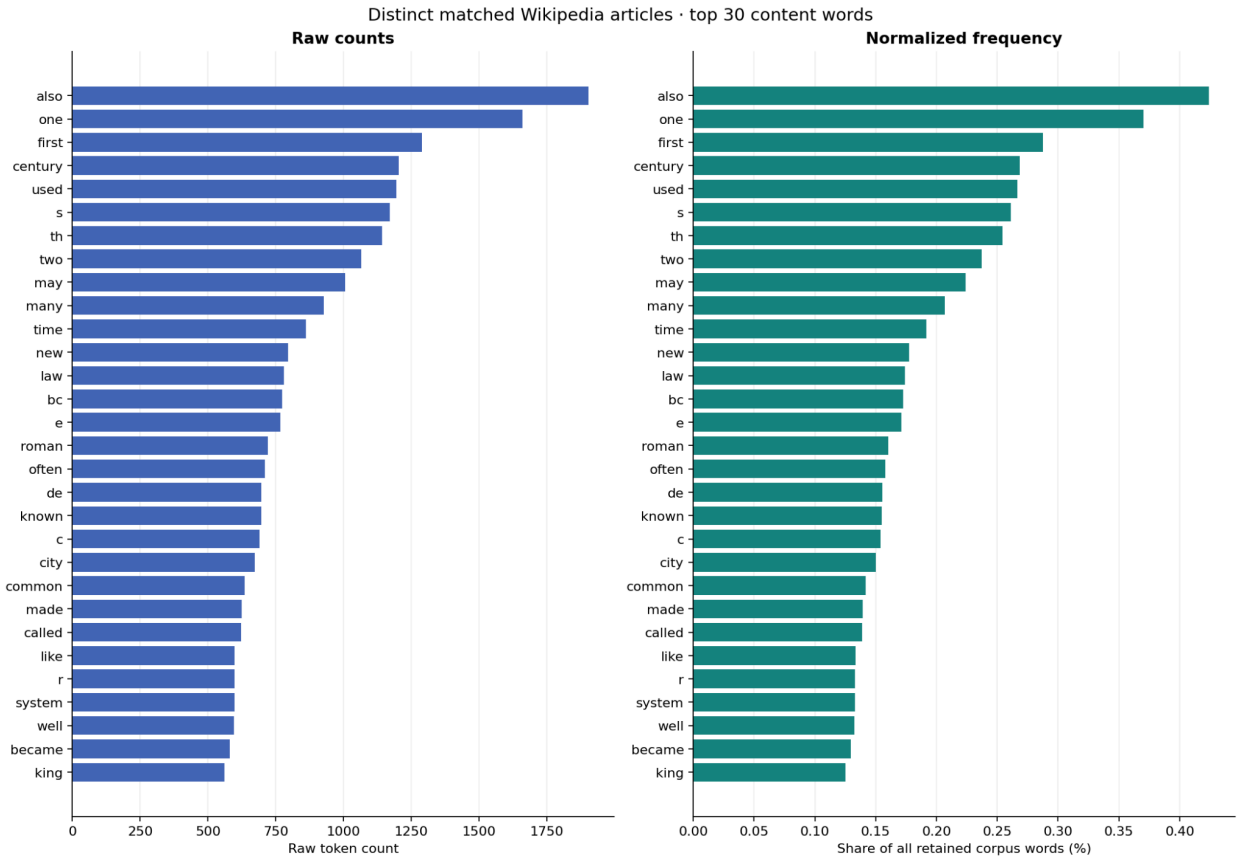
Example BERTopic workflow



Public resources: See the BERTopic project and the starter overview. The diagram shows one possible comparison method. Its optional Super VL topic-refinement stage does not replace the required Hermes agent.

Example word frequency comparisons





The charts illustrate raw counts and normalized frequencies for one sample. They are examples, not target scores for your submission.

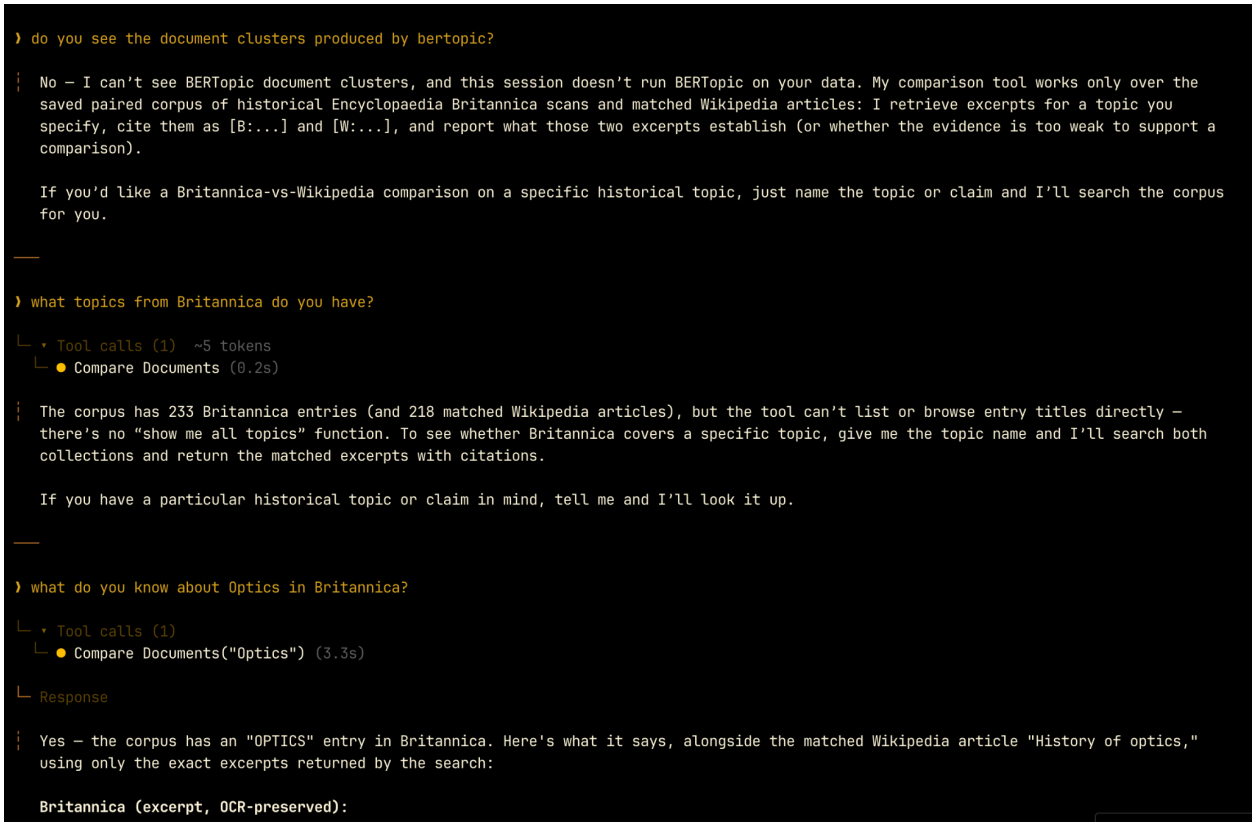
4. Build a comparison agent with Hermes



- Use Nemotron 3.5 Super VL with Hermes to answer questions that compare the two datasets. Document three successful cases and three clear failure cases from your own runs.
- Show when the agent can produce a supported answer and when it cannot. Follow the public Hermes setup guide.

5. Extending the Agent

Experiment customizing the functionality of Hermes. Inspect its capabilities and limitations.



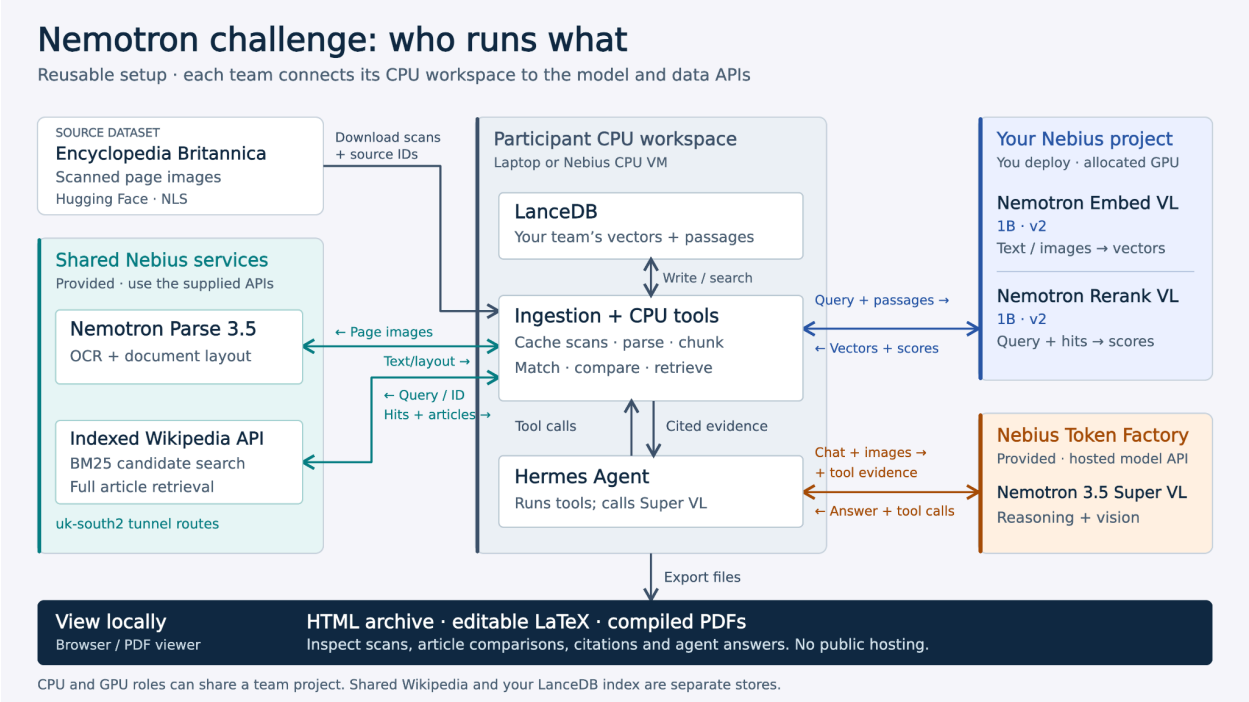
Required models

- Nemotron 3.5 Super VL (nvidia/Nemotron-3.5-Super-VL): Access the model through Nebius Token Factory.
- Nemotron Parse 3.5 (nvidia/nemotron-3.5-parse): Use the shared organizer endpoint listed in the starter configuration.
- Nemotron Embed VL (nvidia/llama-nemotron-embed-vl-1b-v2) and Nemotron Rerank VL (nvidia/llama-nemotron-rerank-vl-1b-v2): Deploy both models on Nebius AI Cloud with the public Docker build recipe.

The Docker recipe serves the embedder and reranker from one image. Parse uses the shared endpoint; Super VL runs through Token Factory. The public recipe pins model revisions for reproducible deployment.

Infrastructure

- Nebius AI Cloud: Run your team's embedder and reranker.
- Nebius Token Factory: Access Nemotron 3.5 Super VL. You can run Hermes and your tools on a laptop or CPU VM.



Deliverables

- A 3-minute video showing successful and failed cases from your pipeline.
- A GitHub repository containing your solution, static website, compiled LaTeX documents, comparison analysis, and three success cases and three failure cases.

Judging rubric (0–100 points)

Assess the submitted repository and 3-minute demonstration video. Add the six category scores below to obtain a total from 0 to 100.

Criterion	Maximum points
Completion of specs	20
Creativity	20
Use of Nemotron Parse 3.5	15
Use of Nemotron Embedder and Reranker	15
Extension of Hermes Agent	10

Use of Nemotron 3.5 Super VL	20
Total	**100**

Scoring guidance

Completion checks use 0 points for missing or unverified work, 2 for partial completion, and 4 when every listed element is demonstrated. Creativity components use whole-point scores from 0 to 10: 0 for no demonstrated contribution, 5 for a promising idea with limited execution, and 10 for a well-executed contribution supported by examples.
Each model and Hermes check uses whole-point scores from 0 to 5: 0 for absent or unverified work, 2–3 for a working but partial result, and 5 for a useful, integrated result supported by examples or execution evidence. Judges may use intermediate scores to reflect the demonstrated quality. An absent or unverified component earns 0 on its corresponding checks; retain points earned on other checks.
Completion assesses coverage of the stated requirements. Model criteria assess implementation quality and contribution. Creativity assesses originality and value; the Hermes extension criterion assesses whether added functionality works and helps.

1. Completion of specs: 20 points

Five checks, each worth 4 points:

- Britannica extraction and archive: extract content from the selected scans and present it as compiled LaTeX documents on a static website.
- Matching coverage: process a sample of 100 selected Britannica pages and match their entries to Wikipedia articles. The page count measures coverage; it does not require 100% matching accuracy.
- Comparison analysis: compare matched sources, explain the method, and cite evidence from both Britannica and Wikipedia.
- Comparison agent and cases: deliver the Hermes comparison agent and document three successful cases and three clear failure cases from the team's own runs.
- Submission: provide the GitHub repository and a 3-minute video showing successful and failed cases. Include the solution, website, compiled LaTeX documents, comparison analysis, and case documentation in the repository.

2. Creativity: 20 points

- Originality 10 points: demonstrate a distinct team contribution beyond the starter, such as a matching method, comparison approach, agent capability, or visualization.
- Value 10 points: demonstrate how the contribution helps users explore or understand differences between historical Britannica and Wikipedia.

3. Use of Nemotron Parse 3.5: 15 points

- Integration 5 points: Parse runs on the selected scans and its output feeds the extraction workflow.
- Extraction fidelity 5 points: extracted content matches the source scans, with errors identified.
- Structure preservation 5 points: reading order, headings, and other relevant page structure carry through into the LaTeX output.

4. Use of Nemotron Embedder and Reranker: 15 points

- Embedder 5 points: demonstrate working use in representing or retrieving relevant content.
- Reranker 5 points: demonstrate working use in rescoring candidates, with the ranking used downstream.
- Combined workflow 5 points: demonstrate how both stages contribute to Britannica-to-Wikipedia matching, using examples. Full credit requires both models to work together.

5. Extension of Hermes Agent: 10 points

- Working extension 5 points: a team-built tool, skill, or workflow adds functionality beyond the starter.
- Demonstrated benefit 5 points: show end to end how the extension helps the comparison agent complete a useful task.

6. Use of Nemotron 3.5 Super VL: 20 points

- Integration 5 points: the model powers the Hermes comparison agent in a working workflow.
- Comparison quality 5 points: answers identify and explain relevant similarities and differences.
- Grounding 5 points: claims are supported by evidence from both sources.
- Failure handling 5 points: the agent identifies missing evidence, uncertainty, or unsupported conclusions and responds appropriately.

Before you start

- Create separate Nebius AI Cloud and Nebius Token Factory accounts. You need access to GPU resources in your Nebius project and a Token Factory API key.
- If you use a Hugging Face token to download the dataset, give it read access. Keep all API keys and tokens out of your repository.

The starter includes a fixed sample and general submission guidance. For this challenge, follow the page-selection and 3-minute video requirements stated in this brief.

Public starter code

Create your team's repository from the public GitHub template. Follow its README and development milestones. The starter contains unfinished stages for your team to implement.

References

- Nebius MCP Server
- NVIDIA Nemotron 3 Embed Ranks #1 Overall on RTEB, Advancing Agentic Retrieval
- BERTopic documentation
- BERTopic for Topic Modeling: Maarten Grootendorst interview
- NVIDIA topic-modelling playbook and public source code
- Accelerate Clustering Algorithms to Achieve the Highest Performance: Allison Ding